

James Philip Iddon

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Experienced applied physicist / systems engineer with 8+ years in the integration, verification, and operation of large-scale, safety-critical silicon particle detectors and their associated data acquisition systems within large, international collaborations. Proven track record in system-level verification and validation (V&V), requirements-driven development, cross-functional team leadership, and issue resolution under mission-critical conditions.

Skills

Technical: Python • C++ • Hardware-software integration • Verification and Validation (V&V) • CI/CD pipelines • Docker • Bash • Git • Linux • Automated testing and deployment • REST API • UART • I2C

Systems Engineering: Systems life-cycle development, requirements analysis, integration, validation, deployment, and operational support • Configuration management and quality assurance

Communication & Coordination: Jira • International stakeholder engagement • Cross-functional team leadership • DevOps operations coordination

Code Quality: Code review workflows • Static analysis tools • Testing automation and continuous integration

Professional Experience

Severo Ochoa Postdoctoral Fellow

IFIC / Universitat de Valencia, Valencia, ES

Apr 2026 – Present

Applied Physicist (Staff, formerly Senior Fellow)

CERN, Geneva, CH

Jul 2022 – Dec 2025

- Planned V&V for electrical characterisation of safety-critical silicon sensor array (ATLAS ITk Pixel outer barrel subsystem, 50-institution project).
- Deployed and maintained over 5 automated test setups for silicon sensor module qualification.
- Managed 24/7 operations and DevOps team for high-reliability sensor system supporting particle accelerator data acquisition at LHC.
- Provided on-call operational and DAQ expertise for 24/7 mission-critical operations over 2 years.
- Developed/maintained DAQ software (C++/Python, CI/CD pipelines) for continuous operations in a mission-critical environment.
- Mentored and led 24/7 shift teams drawn from 300-person international pool.
- Chaired international meetings aligning cross-functional teams over 9 time zones.
- Represented ~30 person operations team in 6000+ member international collaboration.
- Authored 3 novel software packages (Python) including well used operational tools deployed across international collaborations.
- Served as system operations coordinator for record-setting accelerator year.

Applied Physicist (Doctorant)

University of Liverpool / CERN

Oct 2017 – Jul 2022

- Constructed, characterised, and verified 25% of the Outer Barrel of the ALICE ITS2, the largest monolithic pixel sensor array ever built, with 98% manufacturing yield.
- Developed verification software integrated with production readout and V&V strategies for full detector system (12.5 billion pixels, 10m² silicon).
- Validated performance of complete sensor system against requirements.
- Made first measurement of detection efficiency using 5 million cosmic muon tracks across full detector system.
- Presented technical work in international collaboration meetings (170+ institutions) from small working groups to large plenaries, as well as national and international conferences.

Education

Intensive Spanish Course

Taronja Spanish School, Valencia

Dec 2025 - May 2026

PhD in Applied Physics

University of Liverpool / CERN

Oct 2017 – Jul 2022

- Thesis: Construction, Commissioning and Performance Measurements of the Inner Tracking System Upgrade of ALICE at the LHC, focused on advanced systems engineering, integration, and comprehensive verification and validation of complex detector systems.
- Defended June 2022. Shortlisted for the ALICE thesis award.

MPHYS Physics

University of Liverpool

Sep 2013 – Jul 2017

- Integrated masters, specialised in detector physics • Grade: First Class
- Masters project: 'Inner Tracking System Upgrade of the ALICE Experiment at the LHC', characterisation of silicon CMOS MAPS chips.

Full driving licence • Spanish residence permit • English native speaker • Spanish A2+/B1